Extended Abstract

Motivation Video editing plays a crucial role in shaping how viewers interpret actions and infer intent. While prior work in inverse planning has modeled how humans infer agents' goals from observed behaviors, these models typically assume full observability and ignore the influence of editorial choices. Yet, in storytelling and communication, what is shown—and what is omitted—deeply affects interpretation. We address this gap by extending inverse planning to account for spatial edits such as camera cuts and framings, enabling computational reasoning about belief manipulation through editing.

Method Our framework combines rational inverse planning with a model of partial visual observations. We treat spatial edits as probabilistic functions over goal relevance, allowing us to simulate how cuts and close-ups shape viewer inference. Our inverse planner marginalizes over all agent trajectories consistent with observed (edited) frames and estimates audience belief updates over time.

Implementation We instantiate this framework in a 3×5 gridworld where a *baby* agent moves toward either a cake or candy. Editors can apply three camera framings: a full view, a close-up on the cake, or a close-up on the candy. Our inverse planner simulates audience beliefs over time given partial observations, modulated by two key parameters: β (agent rationality) and ζ (framing influence). We compute exact posteriors over latent paths using forward enumeration.

Results We present two qualitative scenarios to illustrate how different framing choices alter viewer beliefs. In one, a cut to the candy followed by the baby's appearance reinforces belief in that goal. In another, absence of the baby leads to belief shifts depending on the strength of framing bias (ζ). We further demonstrate our framework in a POMDP-based MiniGrid environment to show its generalizability to richer perceptual settings and its applicability with an inverse inverse planning interface.

Discussion Our model highlights how editorial framing can serve as a communicative tool, influencing audience inferences alongside agent actions. While our current implementation focuses on spatial edits in a simplified world, it opens avenues for intent-aware editing tools and belief-based narrative analysis. Limitations include assumptions about viewer knowledge and the simplification of editing language.

Conclusion We introduce a novel extension of inverse planning that integrates editorial framing, enabling simulation and optimization of audience belief trajectories. This framework not only explains how edits affect interpretation but also supports inverse inverse planning—selecting actions and edits to achieve desired audience beliefs. Our approach provides a stepping stone toward computational models of narrative communication and inference-aware storytelling tools.

How do directors plan their films?

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Abstract

We propose a computational framework that extends inverse planning models to account for editorial choices in narrative video. While traditional inverse planning explains how viewers infer an agent's goals from observed actions, our approach incorporates partial visual observations caused by spatial editing, such as camera cuts and close-ups. By modeling how different framings influence audience belief updates, we show that editing plays a critical role in shaping interpretation, not just revealing events. We implement this extended inverse planning model in a 2D-gridworld environment and demonstrate its ability to reason over both hidden actions and editorial emphasis. As an application, we introduce inverse inverse planning: selecting agent actions and edits to induce a desired audience belief trajectory. Together, our results highlight how integrating viewer inference into editing models opens up new directions for intent-aware storytelling tools.

1 Introduction

Effective communication of intent is critical not only in storytelling, but also in domains such as human-robot interaction and education. Effective communication of intent is central to storytelling, as well as to domains like human-robot interaction and education. In narrative film, video editing is a powerful tool for shaping interpretation: by selecting which actions to show, conceal, or emphasize, editors can steer the audience's inferences about characters' goals and intentions. Despite its expressive potential, the computational modeling of editing as a means of belief manipulation remains largely unexplored.

We build on the framework of inverse planning, a well-established model of human goal inference that assumes observers interpret behavior by inverting a rational planning process. Prior work has shown that humans can infer agents' goals and beliefs from their actions using such models, and this theory has informed research in cognitive science Baker et al. (2017) and human-robot interaction Dragan et al. (2013). However, these models often assume full observability of actions and environments, overlooking how partial information—especially due to editorial framing—shapes inference.

Motivated by this gap, we propose a framework that extends inverse planning to account for partial visual observations induced by spatial editing. When a viewer sees only part of a scene—due to a close-up or a cut—their beliefs evolve not just from what is shown, but also from what is omitted and how it is framed. Our model integrates both agent behavior and editorial decisions, treating editing as a probabilistic function that influences audience beliefs. This extension allows us to reason about viewer inference under more realistic conditions and opens up new ways to analyze and support narrative communication.

Our work is informed by longstanding insights from film theory and analytical philosophy, which have described how viewers extract meaning from editorial structure. For instance, Greenberg's theory of spatial coherence in narrative film analyzes how cuts preserve or disrupt mental models of space and intention Cumming et al. (2017, 2021). Similarly, phenomena like the Kuleshov effect and

Chekhov's gun reveal how juxtaposition and prominence shape expectation. We aim to operationalize these ideas computationally through belief modeling.

Finally, we present a potential application: inverse inverse planning, where both agent behaviors and editing choices are optimized to induce a desired trajectory of audience beliefs. This reframes storytelling and directing as planning over belief dynamics. We demonstrate our approach through qualitative examples in gridworld environments and propose directions for future work in richer domains. Our results suggest that inverse planning models, extended to account for editing, can support intent-aware storytelling tools and offer a deeper understanding of how viewers make sense of narrative.

2 Related Work

2.1 Computational video editing

Recent advancements in computational video editing have explored automating editing decisions through various methodologies Soe (2021); Wen et al. (2022). Optimization-based approaches Galvane et al. (2015); Merabti et al. (2016), utilize cost functions to enforce cinematic constraints, enabling automatic editing of 3D animations by minimizing continuity errors and optimizing shot composition. Leake et al. introduced a virtual director system employing Hidden Markov Models to select camera shots that align with predefined editing idioms in dialogue-driven scenes Leake et al. (2017). These methods demonstrate the potential of formalizing editing rules to automate the editing process effectively. Incorporating machine learning, recent works have leveraged reinforcement learning to model editing decisions. For instance, a two-stage scheme has been proposed where a virtual editor, trained via reinforcement learning, makes sequential editing decisions to produce videos Hu et al. (2023). Additionally, the integration of large language models (LLMs) into video editing tools, as seen in systems like LAVE Wang et al. (2024), facilitates natural language interactions, allowing users to guide the editing process through conversational interfaces. These approaches highlight the trend towards more intuitive and intelligent video editing systems.

In contrast, our method does not depend on extensive datasets or low-level heuristics. Instead, we leverage the content itself and its impact on the audience to guide editing decisions. By integrating audience inference into the editing process, we aim to, not only select, but also optionally generate content that aligns with desired viewer interpretations, moving beyond traditional rule-based systems.

2.2 Inverse planning

Inverse planning has been widely used as a model of human inference, treating goal and intention recognition as the inversion of rational planning processes Baker et al. (2007, 2009). It is closely related to inverse reinforcement learning, where reward functions are recovered from observed behavior Arora and Doshi (2021). Beyond optimal agents, recent work has modeled agents as boundedly rational planners, capable of suboptimal or noisy behavior, leading to more accurate inferences about human actions Zhi-Xuan et al. (2020). Inverse planning has also been extended to model human interpretations of continuous 3D motions Qian et al. (2021), interventions on others' emotions Chen et al. (2024), and belief-directed communication Goodman and Frank (2016); Ho et al. (2021). Applications in robotics include generating legible and predictable robot motion that aligns with human inference processes Dragan et al. (2013), as well as efforts to endow robots with theory of mind capabilities to better collaborate with humans Gurney and Pynadath (2022). Closer to our work, Chandra et al. have introduced the notion of inverse inverse planning for storytelling, where actions are chosen not to achieve an agent's goals but to manipulate the inferences of a simulated observer Chandra et al. (2023b, 2024).

Our method builds on this foundation but differs in key ways: we use Deep Q-Learning to model agent behaviors under partial observability and apply dynamic programming to select action sequences that optimize audience inference in the context of video editing, a domain that has not been addressed in prior inverse inverse planning work.

3 Method

3.1 Inverse planning

Setup. We begin by considering a simplified scenario, in which a *baby* agent navigates a 2D maze represented by a grid. At each time step t, the agent occupies one grid cell and can deterministically move in any of the four cardinal directions (up, down, left, right) or remain stationary. The agent's objective can only be to reach one of two target objects placed within the grid: a cake or a candy, denoted as $G \in \{G_{cake}, G_{candy}\}$.

Planning. We formalize this scenario as a Markov Decision Process (MDP) defined by the tuple (S, A, T, R, γ), where S is the set of grid cells (states), A is the set of available actions (cardinal moves or stay), $T : S \times A \to S$ is the deterministic transition function, $R : S \times A \times G \to \mathbb{R}$ is the reward function dependent on the chosen goal, and $\gamma \in [0, 1]$ is the discount factor. Specifically, the reward function is designed to provide a positive reward for staying at the goal state.

We solve this MDP using standard Q-learning (**?**) to derive the optimal value function $Q_G(s, a)$ for each goal. We model the agent's policy using a Boltzmann softmax policy with a temperature parameter β:

$$\pi_G(a|s) = P(a|G, s) = \frac{\exp(\beta \cdot Q_G(s, a))}{\sum_{a'} \exp(\beta \cdot Q_G(s, a'))}. \tag{1}$$

Inverse planning. Given a sequence of observed actions $a_{1:t}$ and states $s_{1:t}$, inverse planning involves inferring the likelihood of each possible goal G Baker et al. (2007); Ullman et al. (2009). Formally, we wish to compute the posterior distribution:

$$P(G|s_{1:t}, a_{1:t}) \propto P(a_{1:t}|s_{1:t}, G)P(G), \tag{2}$$

assuming a uniform prior $P(G)$. Under the assumption of Markovian and independent actions conditioned on states, this can be factorized as:

$$P(G|s_{1:t}, a_{1:t}) \propto \prod_{\tau=1}^{t} \pi_G(a_\tau|s_\tau). \tag{3}$$

Each action likelihood is given directly by the Boltzmann policy of the agent, thus:

$$P(G|s_{1:t}, a_{1:t}) \propto \prod_{\tau=1}^{t} \frac{\exp(\beta \cdot Q_G(s_\tau, a_\tau))}{\sum_{a'} \exp(\beta \cdot Q_G(s_\tau, a'))}. \tag{4}$$

3.2 Reasoning over cuts

Partial observability. To extend the inverse planning framework into the domain of video editing. As for now, we only consider spatial edits, leaving temporal edits for further exploration. Rather than showing the entire scene, editors routinely use framing techniques—such as cropping, zooming, or selecting a specific camera angle—restricting what the audience can see. A direct comparison can be found in live broadcasts or multi-camera settings where directors dynamically switch between camera feeds to highlight or conceal key information. We formalize these editorial decisions with a partial observation model. At each time step t, the viewer receives a partial observation $\hat{o}_t = \hat{o}(s_t, a_t)$ derived from the full underlying state and action (s_t, a_t).

We thus define the posterior probability of a goal given the sequence of partial observations up to time t as:

$$P(G \mid \hat{o}_{1:t}) \propto P(G) \cdot P(\hat{o}_{1:t} \mid G) \tag{5}$$

To compute the likelihood term $P(\hat{o}_{1:t} \mid G)$, we marginalize over the set of plausible latent trajectories $\pi(\hat{o}_{1:t})$ that could have generated the observed sequence, assuming the agent acts rationally under goal G:

$$P(\hat{o}_{1:t} \mid G) = \sum_{(s_{1:t}, a_{1:t}) \in \pi(\hat{o}_{1:t})} \prod_{\tau=1}^{t} P(\hat{o}_\tau \mid s_\tau, a_\tau, G) \cdot \pi_G(a_\tau|s_\tau) \cdot P(s_\tau \mid s_{\tau-1}, a_{\tau-1}) \tag{6}$$

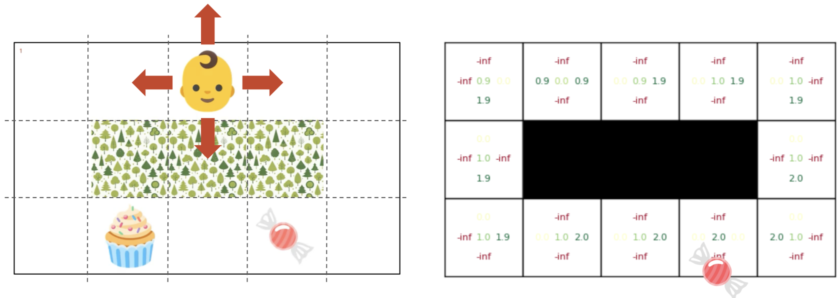

Figure 1: (Left) Our setup environment: a *baby* agent can reach two potential goals. (Right) Q-values for candy as goal

The editing language. In narrative films, editorial framing is rarely neutral. It reflects intentional choices grounded in a shared cinematic language. Selective exposure plays a critical narrative role: close-up shots emphasize a character's intent or emotional salience, while wide shots maintain ambiguity or convey spatial context. These choices are shaped by conventions that viewers intuitively understand, as illustrated by phenomena such as the *Kuleshov effect* (where facial expressions are interpreted differently based on intercut context) and *Chekhov's gun* (where visual prominence implies future relevance).

To model this editorial influence within our framework, we treat framing decisions as probabilistic functions of the goal. Specifically, we define the likelihood of observing a particular partial view $\hat{o}_\tau$ given the true state-action pair (s_τ, a_τ) and the agent's goal G as:

$$P(\hat{o}_\tau \mid s_\tau, a_\tau, G) \propto \exp(\zeta \cdot \text{size}(G)),$$

where size(G) denotes the relative area of the goal in the frame—that is, the proportion of the frame occupied by the goal object—, and ζ is a temperature parameter that modulates how strongly framing influences inferences. This formulation captures how editors shape audience beliefs through spatial composition, allowing our inverse planning model to reason about not only agent behavior but also communicative choices made in the edit.

4 Experimental Setup

Task. We evaluate our inverse planning framework in a custom 3×5 gridworld environment designed to simulate a minimal storytelling setup. In this environment, a *baby* agent navigates toward one of two possible goals: a cake or a candy, while avoiding a central obstacle. The agent receives a reward $r = +1$ upon reaching the goal, and actions are deterministic with a discount factor of $\gamma = 0.99$. Q-values are precomputed using standard value iteration and shown in Figure 1 for reference.

Framing. To simulate editorial framing, we define three types of spatial cuts corresponding to different camera perspectives:

- **Wide shot**: shows the full 3×5 field of view around the agent, including both goals and the agent's actions.
- **Close-up on candy**: crops a 2×3 region centered on the candy, excluding the cake from view.
- **Close-up on cake**: crops a 2×3 region centered on the cake, excluding the candy from view.

Latent paths. Since the environment considered is relatively small, we compute the exact posteriors with the full marginal over paths $\pi(\hat{o}_{1:t})$ using forward simulation via recursive enumeration. Specifically, for each candidate goal G, we exhaustively generate all possible trajectories $(s_{1:t}, a_{1:t})$ that are consistent with the observed sequence $\hat{o}_{1:t}$. The overall likelihood $P(\hat{o}_{1:t} \mid G)$ is then computed as the sum of likelihoods across all consistent paths.

While tractable in our controlled setup, this approach would not scale to larger environments. In more complex scenarios, we would need to rely on approximate inference techniques such as Monte Carlo sampling to efficiently estimate this marginal without exhaustive enumeration Chandra et al. (2023a).

Baselines. To evaluate the added value of modeling editing language, we compare three models:

- **A. Baseline**: Traditional inverse planning, assuming the viewer only updates their beliefs when an action is observed and does not consider unseen actions.
- **B. Literal observations (no editing language)**: An inverse planner reasoning over partial observations, reconstructing possible paths when the agent is out of frame.
- **C. Our full approach**: Our proposed inverse planner reasons over partial observations and interprets cuts as intentional communicative acts.

Evaluation Metric. We use the model's belief trajectory $P(G \mid \hat{o}_{1:t})$ to evaluate how well each model aligns with intuitive audience inferences. While no human ground truth is yet available, we use our own qualitative belief trajectories as our primary metric to:

- Compare belief dynamics across different framing choices and inference models
- Demonstrate how our model captures communicative intent through framing

For future quantitative evaluation, we propose fitting parameters to human annotations and using metrics like KL divergence from annotated belief states.

Inverse planning parameters. Our model includes two key parameters that modulate inference behavior:

- β (**rationality coefficient**): controls the sharpness of the agent's softmax policy. A higher β implies more deterministic, goal-directed behavior, while a lower β reflects greater stochasticity. Results of a vanilla inverse planner in our setup are shown in Figure 2.
- ζ (**framing emphasis coefficient**): modulates how strongly framing (i.e., goal size in the frame) influences the inverse planner. A higher ζ increases the impact of close-ups on inference.

In the next section, the parameters are selected heuristically for illustrative purposes. In future work, they could be fit to human judgment data to prove the validity of our modeling and better align the model with actual audience inferences.

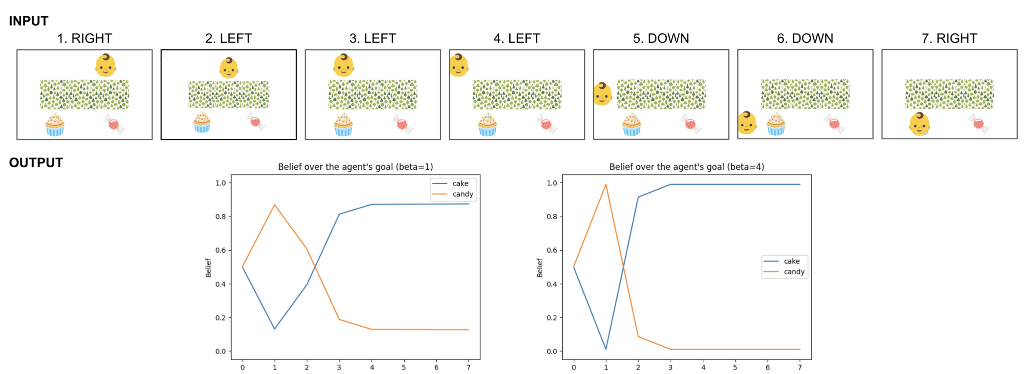

Figure 2: An inverse planner takes a sequence of actions as input and outputs a sequence of corresponding beliefs over goals.

5 Results

We provide qualitative results on two illustrative scenarios designed to probe how our model interprets edited sequences. Rather than quantitative accuracy, which would require extensive experiments with human participants, our focus is on whether the model's belief trajectories align with intuitive audience inferences. In each case, we visualize the sequence of edited frames alongside inferred goal probabilities over time under the compared models.

5.1 Scenario 1

In the first scenario, the sequence begins with a close-up on the candy, hiding the agent's position and motion. After two time steps, the baby enters the frame and begins moving toward the candy. Intuitively, the initial close-up signals narrative relevance of the candy. When the baby appears and approaches the candy, the viewer interprets this as goal-directed behavior, retroactively confirming the editorial cue.

Model A, which lacks inference over unobserved actions, struggles to maintain any consistent belief during the occluded period. Model B improves over A by inferring plausible hidden actions but fails to anticipate the impact of framing. Only Model C anticipates the candy's relevance early on due to the editor's emphasis, and reinforces this belief once the agent is observed. The resulting belief curve matches shown in Figure 3 our narrative expectations.

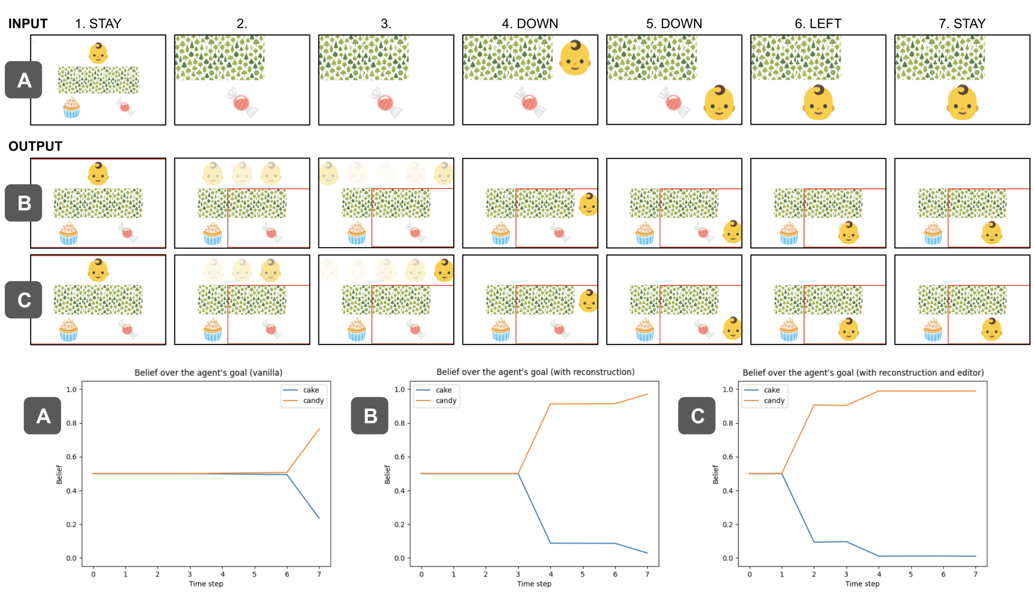

Figure 3: While the vanilla model fails at predicting the agent's goal and trajectory when unseen (A), our inverse planner reasons over unseen actions (B) and editing decisions (C) to determine if the edit is about a *baby* reaching for a cake or a candy.

5.2 Scenario 2

In the second scenario, the editor maintains a close-up on the candy throughout the sequence, and the baby never appears again. This creates a tension: the framing implies that the candy is important, yet the absence of the baby could suggest it went to the cake instead. From a human perspective, both hypotheses are plausible, and belief should shift based on how strongly we trust editorial cues.

Model B, which does not model framing, eventually shifts belief toward the cake due to the lack of confirming visual evidence for the candy. Model C captures the initial bias introduced by the editor's sustained emphasis on the candy. When $\zeta = 1.0$ (C1), the model exhibits an initial tilt toward the candy, which gradually diminishes as the baby fails to appear. With a higher emphasis coefficient ($\zeta = 3.0$ in C2), the belief remains biased toward the candy throughout. These variations shown in Figure 4 illustrate how editorial framing can compete with perceptual cues over time.

Together, these scenarios highlight how our model not only accounts for spatial occlusion but also incorporates editorial decisions into its reasoning. While these are qualitative validations, the belief

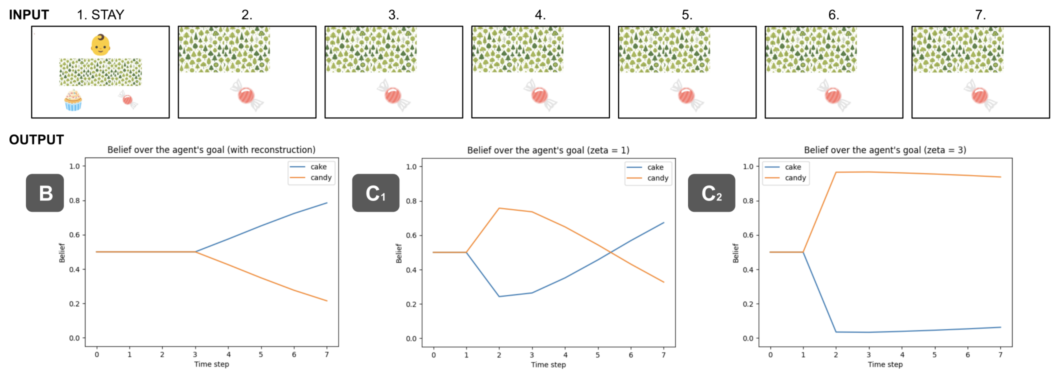

Figure 4: Influence of editorial emphasis on audience belief. Higher ζ sustains belief in the framed object (C2), while lower values allow beliefs to drift as evidence accumulates (C1).

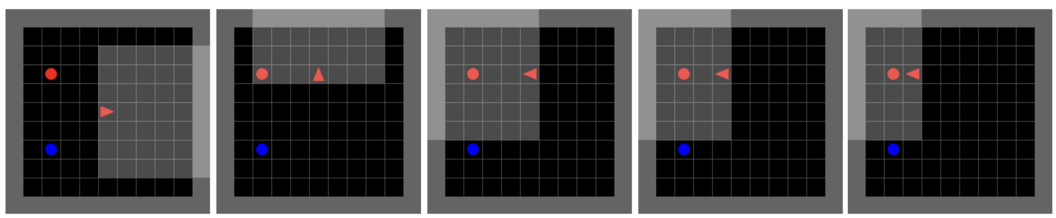

Figure 5: Trained with DQN, the agent finds its way to the red object (goal) by exploring the environment.

curves exhibit patterns that align well with narrative intuitions and demonstrate the potential for modeling audience inference in edited visual sequences.

6 Application

6.1 Generalizability

To demonstrate the generalizability of our framework, we apply it to the `MiniGrid-GoToObject` environment. MiniGrid introduces partial observability, occlusions, and distractor objects—closely mimicking real-world visual ambiguity. Notably, in this scenario, the space of possible inferences grows substantially: due to the agent's limited field of view, reasoning about its intentions now also requires reasoning about what it has or has not seen. As a result, the set of plausible underlying trajectories consistent with an observation becomes larger, introducing additional uncertainty into audience belief modeling. Although we do not implement inverse inverse planning in this setting, we show that the audience inference model could remains applicable: by training a Deep Q-Network (DQN) under a POMDP formulation (Figure 5), we obtain Q-values that allow us to reason about belief inference under more complex perceptual conditions.

6.2 Inverse inverse planning

Film directors make creative decisions to shape the audience's evolving understanding of characters' goals, motivations, and more. These decisions are often informed by imagining how a viewer would interpret the current moment and anticipate what comes next. We formalize this process as *inverse inverse planning*. We select both agent behaviors and edits (i.e., camera framing or cropping) to achieve a **target belief trajectory**—a predefined sequence of audience belief distributions over time.

Objective. Given a desired belief distribution over goals at each time step, denoted $P_{\text{target}}(G, t)$, the goal of inverse inverse planning is to select a sequence of actions and corresponding visual framing $\{(a_t, \hat{o}_t)\}_{t=1}^{T}$ that produce audience observations leading to inferences $P(G \mid \hat{o}_{1:t})$ matching the target. At each timestep t. We optimize:

$$(a_t^*, \hat{o}_t^*) = \arg\min_{a_t, \hat{o}_t} \mathcal{D}_{\mathcal{KL}} \left(P(G \mid \hat{o}_{1:t}), P_{\text{target}}(G, t) \right)$$

where $\mathcal{D}_{KL}$ is the KL-divergence, quantifying how far the audience's predicted belief is from the desired belief.

Implementation. We implement this optimization via dynamic programming. At each step, we simulate audience belief updates using our inverse planning model and recursively choose the action-observation pair that minimizes divergence with the belief trajectory. The result is a policy for generating behavior and cuts that serve specific narrative goals.

Interface. We developed a prototype interface (Figure 6) that allows users to specify target belief trajectories as well as constraints on states or actions at any time step, and observe the resulting agent trajectories and editing decisions. This opens the door to future user-facing tools where creators can "author" audience inferences directly, rather than through manual shot-by-shot construction.

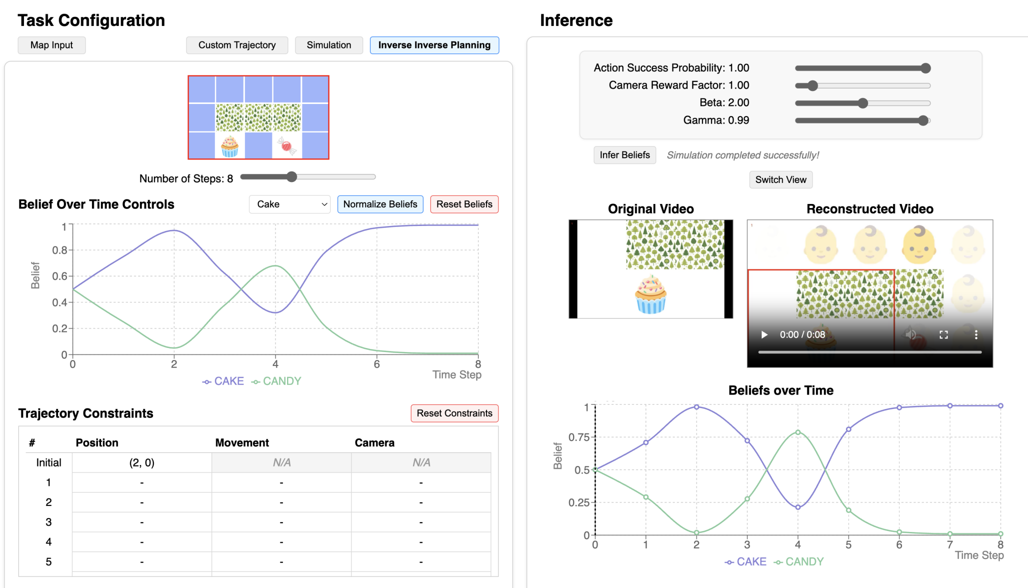

Figure 6: (Left) A user can input a target belief and additional constraints on the desired trajectory. (Right) Our inverse inverse planning model generates a sequence of actions and edits that match the target belief based on our inverse planner.

Future Work. Validating the usability and creative utility of this system would requite empirical user studies with directors, editors, or storytellers that we leave for future studies.

7 Discussion

7.1 Modeling limitations

Our framework relies on strong assumptions: that the audience knows the environment layout, and that the agents space of state, action, or even goals is very limited. It also abstracts away cognitive constraints such as memory limits or perceptual noise, and overlooks the diversity of human priors and interpretations.

The editing model is similarly simplified. We focus only on spatial framing, while real editing involves a much richer language—temporal cuts, juxtaposition, musical cues, etc. For instance, a close-up on the candy followed by one on the cake might suggest conflict or tension, not just two competing goals. These layered narrative cues fall outside the scope of our current formulation.

Despite these limits, our approach offers a first step toward intent-aware editing tools and a tractable platform for modeling viewer inference in a controlled setting.

7.2 Stories as sequences of beliefs

Our approach treats narrative structure as a trajectory of evolving audience beliefs about potential goals. This offers a practical lever for editing, but may fall short of capturing narrative richness, including elements like suspense, humor, or emotional arcs.

A key open question is whether belief trajectories are sufficient or whether creators would prefer to reason at a other levels of abstraction, and if so which ones. Future work should explore how these low-level or high-level models can support more expressive, human-centered storytelling goals.

8 Conclusion

This work presents a computational framework for modeling how editing decisions can shape audience inferences about an agent's goals. By combining inverse planning over an agent's actions with a model of editorial framing, we simulate belief trajectories that viewers might hold under partial observations. We extend this framework to generate agent behaviors and edits that align with target audience beliefs—i.e., *inverse inverse planning*. Our approach demonstrates how simple gridworld setups can serve as controlled testbeds for studying narrative communication and enables potential new tools for inference-aware storytelling. Future work will explore richer editing languages, more realistic environments, and empirical validation with human viewers and creators.

9 Team Contributions

- **Jean-Peïc Chou**: Focused on developing the inverse planning method and action optimization for inverse inverse planning.
- **Saehui Hwang**: Focused on setting up the environments and the inverse planning method.

Changes from Proposal. Our initial plan covered multiple environments (maze, doors-and-keys, help-vs-hinder) and human evaluations. As the project evolved, we narrowed our focus to a single-agent setup (cake vs. candy) and its partially observable variant, allowing us to build a full pipeline and reason carefully about inference and framing. We also introduced a formal model of the editor's communicative role, which was not initially planned.

Team contributions evolved accordingly: while Saehui initially planned to lead multi-agent and evaluation work, she focused instead on setting up environments and co-developing the inference model. Jean-Peïc concentrated on action optimization and inverse planning method. Our original plan was a bit too ambitious. But as we moved from creative ideas about framing to actually building the system, we learned a lot about how storytelling works and enjoyed operationalizing some of its mechanisms.

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